

CDAC settling time 公式比较

目标：把你笔记里的 CDAC settling time 推导，和网上/论文里常见的几种 settling 公式放到同一个框架里比较。本文截图都来自资料原页渲染，不是手画图。

一句话结论

你的公式和网上常见公式不是互相矛盾，而是建模层级不同：

- 常见资料多把 DAC/CDAC 看成黑盒一阶系统，直接写 $V_{err}(t)=V_0e^{-t/\tau}$ ，然后要求误差小于 $1/2$ LSB。
- 你的笔记进一步从 top plate / bottom plate KCL 推出 τ 的来源：底板寄生、顶板寄生、开关电阻共同决定 settling。
- 你得到的 $t_{settle}=\tau \ln x$ 和论文里常见的 $N\tau \ln 2$ 、 $(N+1)\tau \ln 2$ 本质都来自同一个式子，只是最大阶跃幅度 V_0 、精度要求、阵列结构不同。

公式总览

类型	常见公式	适用场景	和你的区别
普通采样电容 settling	$t=-\tau \ln(E)$ ，或按 $1/2$ LSB 推出约 $t \approx (N+1)\tau \ln 2$	输入采样电容、外部源阻抗、ADC tracking phase	关注输入采样，不一定是 SAR 内部 CDAC bit cycling
SAR/CDAC 黑盒误差	$V_{err}(t)=V_0e^{-t/\tau}$	只关心某次 DAC step 是否 settle 到精度内	没展开 τ 来自哪些寄生/开关
redundancy 放宽 settling	前几位只需较低 bit 精度，例如 3-bit/4-bit accurate settling	非二进制/冗余 SAR	通过数字冗余容忍 settling error，而不是严格每步 $1/2$ LSB
你的 KCL 推导	$\tau=RC_u(\alpha_P+\frac{\alpha_T}{1+\alpha_T})$ ， $t \geq \tau \ln x$	split CDAC，考虑 top/bottom parasitic	更像 transistor-level/CDAC-internal model

截图资料

截图 1：普通 switched-capacitor ADC 的 settling time 公式

2. Determining Settling Time

The settling time required for a given application is determined by the ADC input circuit, external circuitry (e.g., anti-alias filter), and the ADC settling time specification. If proper settling requirements are not met, then the ADC may not meet the specifications posted in the data sheet. One must consider the settling time of the ADC input circuit, external circuitry, and the minimum required by the ADC specification in order to calculate settling time requirements. We design to the most restrictive requirement.

2.1 Minimum Settling Time Specification

Silicon Lab's ADC specification requires a 1.5 μ s tracking time. Even though [Figure 1.1 Equivalent ADC Circuit for Estimating Settling Time on page 1](#) presents an equivalent circuit for settling time estimation, the actual ADC peripheral has numerous components that affect settling time such as switched capacitors, transmission gates, etc. Thus, the minimum specified settling time is 1.5 μ s. If the calculated settling time using the equivalent circuit in [Figure 1.1 Equivalent ADC Circuit for Estimating Settling Time on page 1](#) or the external circuit is greater than 1.5 μ s, then the settling time will be dictated by external components.

2.2 Settling Time of the ADC Input Circuit

Because the equivalent input tracking circuit of the ADC is an RC circuit, we will calculate settling time in terms of time constants. It is useful to specify settling time as the number of time constants it will take for an accuracy specified as a fraction of the least significant bit (LSB):

$$\text{LSB} = \frac{V_{\text{REF}}}{2^N}$$

To calculate the time (t) required for the sample capacitor voltage to settle to within one-fourth of an LSB of the input voltage, we derive an equation for the calculation:

$$V(t) = V_{\text{in}} \times \left(1 - e^{-\frac{t}{\tau}}\right)$$

In this equation, V_{in} is the input voltage at the input pin of the device and the time constant is $\tau = RC$. Solving for t in terms of the number of time constants, τ , we obtain the result:

$$t = -\ln\left(1 - \frac{V(t)}{V_{\text{in}}}\right) \times \tau$$

To calculate the voltage to be within 1/4 LSB of the input voltage, assuming a full-scale step input ($V_{\text{in}} = V_{\text{REF}}$):

$$V(t)_{\frac{1}{4}\text{LSB}} = V_{\text{REF}} \times \left(1 - \frac{1}{2^N \times 4}\right)$$

Substituting and again assuming a full-scale step input ($V_{\text{in}} = V_{\text{REF}}$), we obtain the following result:

$$t = -\ln\left(\frac{1}{2^N \times 4}\right) \times \tau$$

In this equation, N is the number of ADC bits. Assuming a 12-bit ADC, this equation becomes:

$$t = -\ln\left(\frac{1}{4096 \times 4}\right) \times \tau = \ln(4096 \times 4) \times \tau = 9.7 \times \tau$$

Note that this equation is for a 12-bit ADC, where a 1/4 LSB accuracy is desired. If we are using an 8-bit ADC and also want 1/8 LSB accuracy, then this equation becomes:

$$t = \ln(256 \times 8) \times \tau = 7.6 \times \tau$$

As shown in [Figure 1.1 Equivalent ADC Circuit for Estimating Settling Time on page 1](#), we estimate the impedance and capacitance to be $R = 5 \text{ k}\Omega$ and $C = 10 \text{ pF}$. Substituting the values of R and C, $\tau = 50 \text{ ns}$. Thus, the settling time for the 12-bit ADC and 1/4 LSB accuracy is about 500 ns, and the 8-bit ADC with 1/8 LSB accuracy is 380 ns. However, the ADC specification for minimum settling time is 1.5 μ s, which is more restrictive in both cases.

A 10-bit SAR ADC With Two Redundant Decisions and Splitted-MSB-Cap DAC Array

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Abstract—A novel switching method is proposed and implemented in a 10-bit fully differential SAR ADC. When compared with the charge recycling or the set-and-down approach, the proposed scheme uses a single-ended switching procedure to reduce the total switching energy by 80% or 20%, respectively. Besides, two redundant decisions are inserted in the SAR ADC with a 4b/4b/4b configuration, which is capable of more than 30% speed enhancement. The 10-bit SAR ADC's performance is verified through Matlab simulation and it achieves 10-b resolution. After a Monte-Carlo analysis (100-times), the mean value of the SNDR is 61.8dB, and the maximum INL and DNL are 0.05LSB and 0.1LSB, respectively.

I. INTRODUCTION

Conventional SAR ADCs use a binary-weighted capacitive array to implement the DAC and the sample-and-hold function [1]. Recently, when compared with the flash and pipeline converters, SAR ADCs have been widely used for low-power applications.

The power consumption of SAR ADCs depend on three dominant circuit components: the digital logic, the comparator and the DAC array. In recent years, work has been focused on coming up with smart switching techniques to decrease the switching power, such as charge-recycling [2] and set-and-down method [3]. The traditional topology wastes a lot of energy on changing the voltage on the top plate of capacitors and switching the next capacitor from ground to V_{ref} . Though the charge-recycling method uses the same amount of capacitance to improve slightly this aspect, it still has the disadvantage of doubling the number of switches. On the other hand, the set-and-down scheme reduces half capacitance and costs much less switching energy, but, it still requires a large sum of energy to convert middle output codes.

Moreover, the speed of the SAR ADC is limited by the settling time of the DAC, SA logic and decision time of the comparator. For a n -bit SAR ADC, it takes n decisions for every conversion, and every decision must be as accurate as the full ADC resolution. The requirement of settling error for every decision can be expressed as:

$$settling\ error = V_0 \times e^{\left(\frac{-t}{\tau}\right)} < V_{ref} / 2^{n+1} \tag{1}$$

With settling time, $t > \tau \times \ln\left(\frac{1}{2^{n+1}} \times \frac{V_{ref}}{V_0}\right)$ \tag{2}

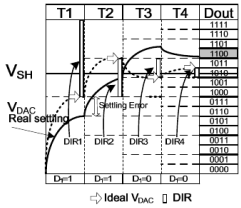


Fig. 1 Example of settling error causing wrong decisions.
where V_0 is the desired settling amplitude in each step.

When the DAC settling time is incomplete the ADC can't export accurate decision codes. As shown in Fig.1, V_{SH} is the sample input and the ideal conversion settling is the dotted line where the white bars are the ideal determined input range (DIR). Since there are some settling errors the output codes don't exhibit n bit accuracy.

Therefore, a new switching scheme is proposed which uses a single-ended switching procedure, not only decreasing the dimension of the capacitor array, but also reducing the energy consumption necessary for converting the middle-scale output codes. Besides, for relaxing the requirement of settling time, redundant decisions are inserted in the SAR ADC. Plus, the addition of digital error correction [4] is also implemented to achieve the desired resolution.

The paper will be organized as follows: Section II introduces the proposed switching scheme and energy saving; Section III reviews the algorithm of redundancy and presents its implementation in the proposed switching technique; Section IV presents the simulation results and the conclusions are drawn in Section V.

II. PROPOSED SWITCHING METHOD

As shown in Fig.2, the DAC array exhibits a new topology that uses 4 unit capacitors to achieve a 3-bit SAR ADC. The proposed switching algorithm applied in this SAR ADC is a single-ended switching procedure, which only switches one capacitor on one side of the capacitive array after the first transition.

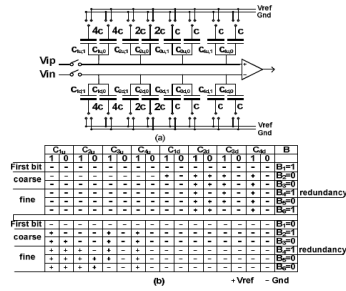


Fig. 7 (a) A 5-bit proposed SAR ADC; (b) corresponding switching table.

However, the operation for this case actually shifts V_{DAC} to the bottom or top of the DIR ($B1=1$ or $B1=0$) rather than to its center. So, it is necessary to switch the up or down second capacitor to ground or V_{ref} , separately, in such a way that V_{DAC} will be moved to the desired level of DIR.

IV. SIMULATION RESULTS OF THE PROPOSED SAR ADC

The 10-b SAR ADC with a 4-b/4-b/4-b configuration has 512 unit capacitors on each side array, which consists of two binary weighted capacitor arrays with total 256 capacitances. It reduces the capacitance to half with respect to the conventional SAR and also implements a common-centroid topology efficiently in the layout of the DAC array.

$$\begin{array}{ccccccc}
 & B1 & B2 & B3 & B4 & & \\
 & B5 & B6 & B7 & B8 & & \\
 + & & & & & B9 & B10 & B11 & B12 \\
 \hline
 D1 & D2 & D3 & D4 & D5 & D6 & D7 & D8 & D9 & D10
 \end{array}$$

Fig.8. Output codes of the proposed SAR ADC

In the topology, two bit redundancies are inserted: $B5$ and $B9$. The first brings in $1/16 V_{ref}$ offset before the first decision and the second is $1/128 V_{ref}$ before the fifth decision. As shown in Fig.8, the digital correction is also applied in the SAR ADC.

Due to the single-ended switching scheme and assuming the same time constant at each step, the first 4b requires 3-b accurate DAC settling: $\tau \ln\left(\frac{1}{2^{k+1}} \cdot \frac{V_{ref}}{V_{ref}/2}\right)$. Then, the second

and final 4b both require 4-b accurate DAC settling. Therefore, the settling time of 4b/4b/4b case for each step is 4 accurate DAC settling. When compared with the set-and-down ADC with 10 accurate DAC settling, the whole DAC settling time is enhanced by more than 30%.

In the Matlab model of the proposed SAR ADC a Gaussian random variable with standard deviation of 1% is added to the value of the unit capacitor. Considering that all conditions are identical, under the condition of the 100-times Monte-Carlo capacitor mismatch simulation, the mean value is 61.8dB SNDR, shown in Fig.9. Finally, Fig.10 depicts the maximum DNL and INL that are 0.1LSB and 0.05LSB, respectively.

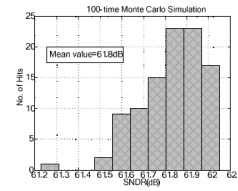


Fig.9. 100-time Monte-Carlo simulation of the proposed SAR ADC.

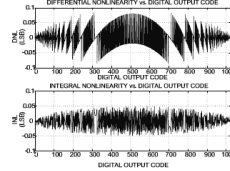


Fig.10. INL and DNL vs. digital output codes.

V. CONCLUSIONS

This paper presents a 10-bit SAR ADC with a single-ended switching procedure and 2-bit redundancy. This switching technique reduces 50 percent of the area of the DAC array and imposes better switching energy efficiency in the converter. Furthermore, the redundancy method enhances the settling speed of the DAC and relaxes the comparator's accuracy.

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(SNDR of 60 dB). Beyond this value, its usefulness is limited by signal dependent charge injection [51]. In the design space of ADCs targeting more than 10 bits of resolution, bottom plate sampling is almost exclusively employed (see e.g. [52], [53] with a very recent exception [54]).

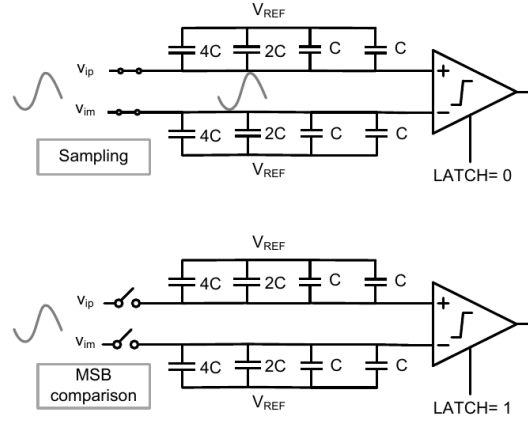


Fig. 3.4 Top plate sampling and MSB comparison.

C. SAR Loop Delay Optimization

A SAR ADC comprises of three main circuit blocks: a comparator, a capacitive DAC and SAR logic as shown in Fig. 3.5. As a result, the speed of a SAR ADC is limited by the sum of the delays through these blocks (see Fig. 3.5): DAC settling (T_{DAC}), comparator decision time (T_C) and SAR logic delay (T_L). Maximizing the speed therefore boils down to a joint minimization of all three components. To illustrate the associated tradeoffs, consider a DAC MSB circuit slice shown in Fig. 3.6. The DAC uses an inverter circuit to drive the bottom plates of the capacitors (connected to V_{REF} during sampling) and the size of the PMOS device is defined by the required on-resistance during sampling. The optimum size for the NMOS follows from a more complex consideration that includes the loading of the SAR

If R_{ref} is made equal to R_{gnd} (in Fig. 3.18), the time constants in the two paths are equal leading to pole-zero cancellation and the input step is directly transferred to the output leading to very fast settling of the DAC output voltage. This holds true for other DAC transitions as well and therefore improves the overall speed of operation. Fig. 3.19 shows the plot of settling time (for 0.5 LSB settling accuracy) vs. the ratio of R_{ref} to R_{gnd} for this ideal model. We see that a 10% mismatch in resistances causes substantial loss in settling time improvement due to a pole-zero doublet in the transfer function. This means that a relatively good matching between pull-up and pull-down resistance is needed to avail the benefits of time-constant matching (also see Fig. 3.21).

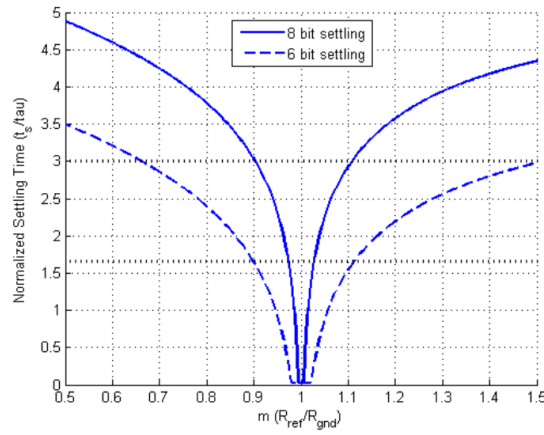


Fig. 3.19 Normalized⁷ settling time vs. resistance ratio for the ideal model. When the time constants are matched, pulse transfer occurs resulting in close to zero settling time.

B. Detailed Analysis

Unfortunately, implementing the switches using MOS transistors results in variable resistance during the DAC switching transient. This section presents the effect of this nonideality on DAC output voltage settling in detail. The switches to V_{ref} (in

⁷ Settling time normalized to time constant (τ) given by $\tau = (R_{ref} + R_{gnd}) \times C_{DAC}/2$, $C_{DAC} = 32C$ (in this example).

From (3.4), we see that using an NMOS transistor for ground switches results in a decaying exponential settling response of the DAC output voltage (instead of an instantaneous step). Equations (3.2) and (3.4) can be numerically solved to calculate the settling times of the DAC output voltage as a function of $R_{\text{ref}}/R_{\text{gnd}}$, as shown in the next section.

C. MATLAB Simulations

For solving (3.2) numerically using MATLAB, the following values were assumed: $V_{\text{DD}} = V_{\text{ref}} = 1\text{ V}$, $C = 128\text{ fF}$ (Total capacitance in the DAC array = 256 fF), $V_{\text{th}} = 0.36\text{ V}$. The ratio of $R_{\text{ref}}/R_{\text{gnd}}$ (variable m) was varied from 0.75 to 1.3. Fig. 3.21 shows the comparison of normalized settling times for 6, 8 and 10 bit settling of the DAC output voltage assuming 0.5 LSB accuracy. The settling times are normalized to the case when $m=1$ ($R_{\text{ref}}=R_{\text{gnd}}$). It can be seen that settling time reduces beyond the case of matched time constants ($m=1$). This is because the zero in the transfer function provides an overshoot, which further improves settling time if the overshoot magnitude is contained within the settling error bound. For 10 bit accuracy in settling, this extra improvement is much less compared to a 6 bit settling, as expected.

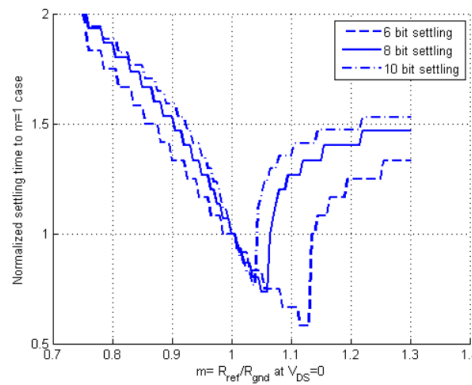


Fig. 3.21 Comparison of normalized settling times (MATLAB simulation) for DAC output voltage settling.

Stanford time-domain DAC switching

来源: [Stanford thesis on high-speed SAR / two-CDAC pipelined

SAR](https://stacks.stanford.edu/file/druid%3Ajc388fn7862/VaibhavT_thesis_final-augmented.pdf)。这份资料强调 SAR loop delay 由 DAC settling、comparator decision、SAR logic delay 组成，并讨论通过 CDAC / switch resistance / pole-zero cancellation 提高 DAC settling speed。

共同母公式

大多数 settling time 公式都能回到这个一阶残差模型：

$$V_{\text{err}}(t) = V_0 e^{-t/\tau}$$

要求最终误差小于允许误差 V_{tol} ：

$$V_0 e^{-t/\tau} \leq V_{\text{tol}}$$

两边取对数：

$$t \geq \tau \ln \left(\frac{V_0}{V_{tol}} \right)$$

所以所有差别都藏在三个量里：

1. τ 怎么定义；
2. 初始误差/阶跃 V_0 多大；
3. 容许误差 V_{tol} 取多少。

普通 $N\tau \ln 2$ 或 $(N+1)\tau \ln 2$ 怎么来的

如果要求 settle 到 $1/2$ LSB：

$$V_{tol} = \frac{V_{FS}}{2^{N+1}}$$

如果最坏初始阶跃大约是满量程：

$$V_0 \approx V_{FS}$$

那么：

$$t \geq \tau \ln \left(\frac{V_{FS}}{V_{FS}/2^{N+1}} \right) = (N+1)\tau \ln 2$$

如果认为最坏 CDAC step 是 $V_{FS}/2$ ，则：

$$t \geq \tau \ln \left(\frac{V_{FS}/2}{V_{FS}/2^{N+1}} \right) = N\tau \ln 2$$

所以网上看到的 $N\tau \ln 2$ 、 $(N+1)\tau \ln 2$ ，主要差在他们把最大 step 取成 $V_{FS}/2$ 还是 V_{FS} 。

你的公式怎么来的

你笔记的电路模型是：

- n 个单位电容 C_u ；
- 顶板节点为 V_{DAC} ；
- 第 k 个底板节点为 V_k ；
- 底板由参考 V_{Rk} 通过开关电阻 R 驱动；
- 顶板寄生 $C_T = n\alpha_{TC_u}$ ；
- 每个底板寄生 $C_P = \alpha_{PC_u}$ 。

1. 顶板 KCL

顶板浮动，因此单位电容电流和顶板寄生电流满足：

$$\sum_{k=1}^n C_u (\dot{V}_{DAC} - \dot{V}_k) = -C_T \dot{V}_{DAC}$$

整理：

$$\sum_{k=1}^n \dot{V}_k = n(1 + \alpha_T) \dot{V}_{DAC}$$

积分：

$$\sum_{k=1}^n V_k = n(1 + \alpha_T) V_{DAC} - M$$

这个式子说明：底板总变化通过电容耦合到顶板，但顶板寄生会稀释顶板变化。

2. 底板 KCL

每个底板节点通过 R 被参考电压驱动，同时要给 C_u 和底板寄生 C_P 充放电：

$$RC_u(\dot{V}_{DAC}-\dot{V}_k)=RC_P\dot{V}_k+(V_k-V_{Rk})$$

对所有 k 求和并代入顶板关系，得到一阶微分方程：

$$\tau \dot{V}_{DAC} + V_{DAC} = \frac{1}{n(1+\alpha_T)} \left(M + \sum_{k=1}^n V_{Rk} \right)$$

其中：

$$\tau = RC_u \left(\alpha_P + \frac{\alpha_T}{1+\alpha_T} \right)$$

这个 τ 很有意思：如果 $\alpha_P=0$ 且 $\alpha_T=0$ ，模型里 $\tau=0$ ，表示理想电容阵列在电荷重分布下瞬时到最终值。真正拖慢它的是寄生电容和有限开关电阻。

3. 指数响应

一阶方程解为：

$$V_{DAC}(t) = V_{DAC}(\infty) + [V_{DAC}(0) - V_{DAC}(\infty)] e^{-t/\tau}$$

也就是：

$$\Delta V_{DAC}(t) = \Delta V_{DAC}(\infty) (1 - e^{-t/\tau})$$

残差误差：

$$V_{err}(t) = \Delta V_{DAC}(\infty) e^{-t/\tau}$$

4. 为什么得到 $t \geq \tau \ln x$

如果一次切换了 x 个单位电容，且底板参考阶跃为 ΔV_X ，则最终顶板变化：

$$\Delta V_{DAC}(\infty) = \frac{x}{n(1+\alpha_T)} \Delta V_X$$

若这里 $\Delta V_X = V_{FS}/2$ ，而有效 $1/2$ LSB 为：

$$V_{tol} = \frac{V_{FS}}{2^{2n(1+\alpha_T)}}$$

代入母公式：

$$\frac{x}{n(1+\alpha_T)} \frac{V_{FS}}{2} e^{-t/\tau} \leq \frac{V_{FS}}{2^{2n(1+\alpha_T)}}$$

公共项全部抵消：

$$x e^{-t/\tau} \leq 1$$

所以：

$$t \geq \tau \ln x$$

如果 split array 的最坏切换单位数是：

$$x_{\max} = 2^{N-2}$$

那么：

$$t_{\text{settle, min}} = \tau \ln(2^{N-2}) = \tau (N-2) \ln 2$$

和网上公式的具体差别

差别 1： τ 的定义

网上常见：

$$\tau = R_{on} C_{eq}$$

你的笔记：

$$\tau = RC_u \left(\alpha_P + \frac{\alpha_T}{1 + \alpha_T} \right)$$

你的 C_{eq} 实际上是：

$$C_{eq} = C_u \left(\alpha_P + \frac{\alpha_T}{1 + \alpha_T} \right)$$

也就是说，你不是把整个 CDAC 总电容都放进时间常数，而是只把会被有限开关电阻充放电的寄生效应放进去了。

差别 2：最大阶跃 V_0

普通公式常用：

$$V_0 = V_{FS}, \quad V_0 = \frac{V_{FS}}{2}$$

你的公式使用：

$$V_0 = \frac{x}{n(1 + \alpha_T)} \frac{V_{FS}}{2}$$

所以你的 settling time 直接和切换单位电容数 x 相关。

差别 3：阵列结构

普通 binary CDAC 的 MSB step 可能对应 $x \approx 2^{N-1}$ 或类似数量级，所以得到接近 $N\tau \ln 2$ 。

你的 split array 假设里：

$$x_{\max} = 2^{N-2}$$

所以得到：

$$(N-2)\tau \ln 2$$

这不是更松或更严的普适结论，而是由 split array 最大切换规模决定的。

差别 4：是否允许 redundancy

冗余 SAR 论文会说：前几步不必每一步都 settle 到完整 N bit 精度，因为后面有数字校正空间。于是 settling requirement 可以从 N -bit accurate 变成 3-bit、4-bit accurate。

你的公式默认没有显式利用 redundancy，而是在每次切换后按 $1/2$ LSB 约束残差。

什么时候该用哪个公式

你在估什么	推荐公式
输入采样网络 settling	Silicon Labs / switched-capacitor ADC 那类 $\tau \ln(V_0/V_{tol})$
粗估 SAR CDAC 每步需要多久	$\tau \approx N\tau \ln 2$ 或 $(N+1)\tau \ln 2$
比较不同 switching scheme 的速度	$\tau \geq \tau \ln x$ ，因为 x 直接体现切换电容数
做 CDAC parasitic-aware 分析	你的 $\tau = RC_u(\alpha_P + \alpha_T/(1 + \alpha_T))$
有 redundancy / non-binary search	按每一段实际需要的 bit accuracy 分段算 settling

需要注意的限制

你的推导很适合解释 CDAC 内部寄生导致的 settling，但它没有包含这些效应：

- reference buffer 输出阻抗和动态压降；
- reference decap 和供电网络；
- 开关电阻随电压变化的非线性；
- comparator input parasitic 与 kickback；

- bottom plate switching 的具体时序重叠；
- differential CDAC 两边的 common-mode settling。

如果用于论文/报告，建议把它表述为：

在忽略 reference driver dynamics、开关非线性和 comparator kickback 的条件下，split CDAC 的一阶 settling time 可由 bottom/top parasitic KCL 推得。

参考来源

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